Workshop 3

1. Consider a survey in which a random sample of vegetarians and non-vegetarians was taken, and for which individual we record weight y_i, height x_i, and gender. A suitable model for the data is

$$\mathbb{E}(y_i) = \alpha + \nu_k + \gamma_j + \beta x_i \text{ if } y_i \text{ is for diet } k \text{ and gender } j, \quad j \in \{1,2\}, \quad k \in \{1,2\},$$

where $k = 1$ for vegetarian and $k = 2$ for non-vegetarian, and $j = 1$ for female and $j = 2$ for male. We assume $y_i \sim N(\mu_i, \sigma^2)$, $i = 1, \ldots, n$, and all Greek letters denote model parameters.

   (a) Write out the model (design) matrix and parameter vector for this model, ignoring identifiability issues For concreteness, assume a random sample of $n = 8$ subjects, the first four vegetarian and the next four non-vegetarian, with the first two in each group female and the next two male.

   (b) Re-write the model matrix and parameter vector in a form that makes the model identifiable.

2. (a) Consider the linear model

$$y_{ij} = \alpha + \beta_i + \gamma_j + \delta z_{ij} + \epsilon_{ij}, \quad i = 1, \ldots, 3, \quad j = 1, \ldots, 4,$$

   where α, β_i, γ_j, and δ are parameters, y_{ij} and z_{ij} are observed variables, and $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$. Write this model in matrix-vector form $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, without worrying about identifiability constraints.

   (b) Rewrite your model from part (a) so that it is identifiable.

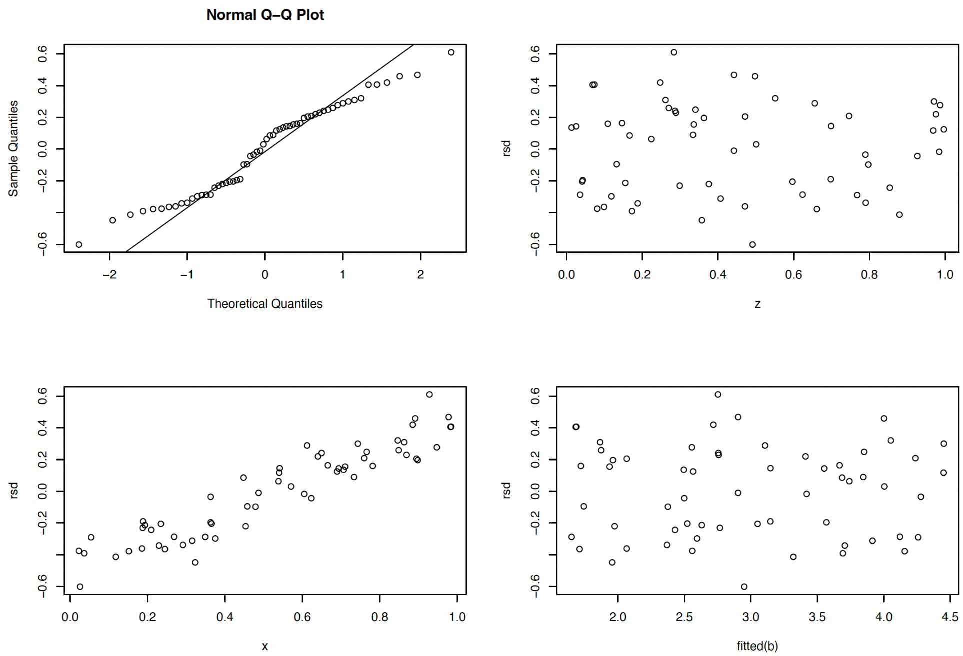
   (c) In the following call to the `R` function `lm`, `a` and `b` are factor variables, and `v` and `w` are continuous variables.

```
lm(y ~ a*b + v + w + I(v*w))
```

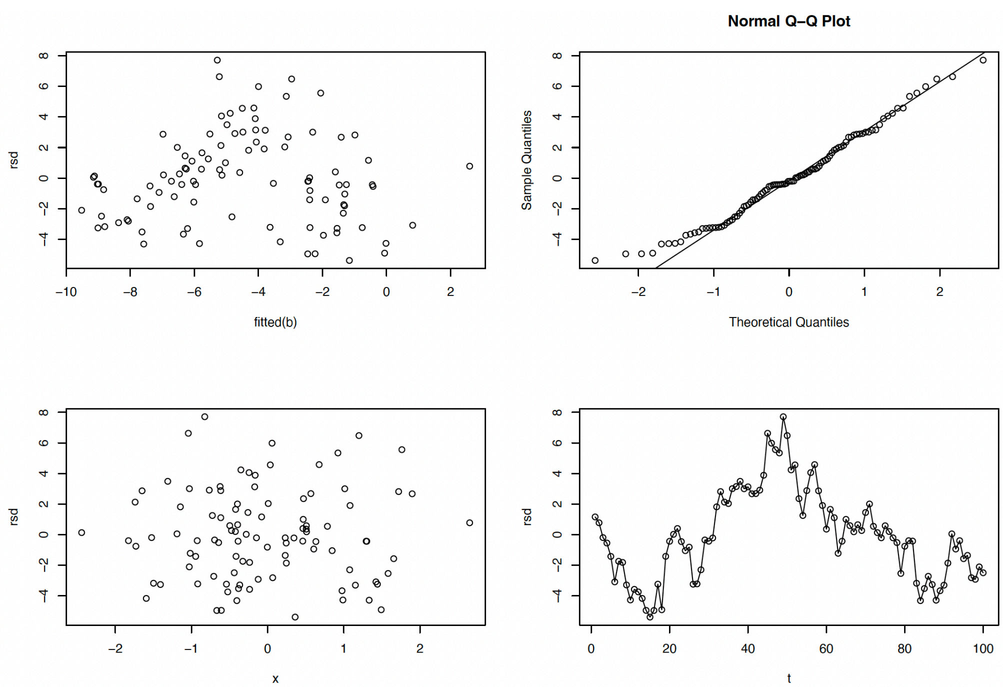
   Write down the mathematical form of the model being fitted, give it as an explicit equation and distributional assumption, but do not write it in matrix-vector form. You may ignore identifiability constraints.

3. In each part of this question, some `R` code is used to fit a linear model to a dataset and to generate the residual plots shown. For each case, describe the main problem (if any) revealed by the plots and, where possible, suggest how it might be addressed.

   (a)
```
names(dat)
"y"   "x"   "z"   "fac"
b <- lm(y ~ fac + z, data = dat)
rsd <- residuals(b)
par(mfrow = c(2,2))
qqnorm(rsd); qqline(rsd)
with(dat, plot(z, rsd))
with(dat, plot(x, rsd))
plot(fitted(b), rsd)
```



(b)
```
b <- lm(y ~ x + t)
rsd <- residuals(b)
par(mfrow = c(2,2))
plot(fitted(b), rsd)
qqnorm(rsd); qqline(rsd)
plot(x, rsd)
plot(t, rsd); lines(t, rsd)
```



4. This question is about model checking, transformation and using factor variables in `R`. You will use the `InsectSprays` dataset supplied with R. The data frame contains `count` values, giving the number of pest insects found on agricultural test plots, each of which was treated with one of six insecticide formulations (the factor variable `spray` indicates which). The aim is to determine whether formulation affects the final insect count, and if so, which formulation performs best.

   (a) Produce a plot of `count` against `spray` type. How would you interpret the plot?

   (b) Fit a linear model with `count` as the response and `spray` as the predictor. Examine the residuals versus fitted values (or, alternatively, a plot of the square root of the absolute value of the standardised residuals against the fitted values). Is the constant-variance assumption satisfied?

   (c) Is the normality assumption reasonable for this linear model?

   (d) Count data are often more naturally modelled using a Poisson distribution rather than a normal distribution. Using the normal model as an approximation to Poisson (or a transformation of Poisson) is typically acceptable if we can correct for the fact that Poisson data do not have constant variance. A variance-stabilising transformation can help. In particular, using the square root of `count` as the response should approximately stabilise the variance. Refit the model using $\sqrt{\texttt{count}}$ as the response. Does this improve matters?

   (e) Look at the `summary` of your transformed model. What is the interpretation of the `(Intercept)` term? What about the coefficients for the other factor levels?

   (f) Sometimes it is helpful to change the reference level of a factor variable. For example, suppose we want to compare all formulations with formulation `C`. As discussed in the lecture notes, this can be achieved using the `relevel` function:

   ```
   InsectSprays$spray <- relevel(InsectSprays$spray, "C")
   ```

   Refit the model and examine the summary output again. How has it changed? What would you now conclude?

   (g) Finally, refit the model to the untransformed counts, again using `C` as the reference level. Compare the summary output with that from the transformed model. Your conclusions should be quite different–why is this, and which set of conclusions would you trust?

5. The *High School and Beyond* dataset is available in `hsb` in the `faraway` package in `R` (you will need to install this package). The data are a subset of the *High School and Beyond* study conducted by the National Education Longitudinal Studies program of the National Center for Education Statistics. The object `hsb` is a data frame with 200 observations and 11 variables. To learn more about the variables, type `help(hsb)`. One goal of the original study was to identify the factors influencing students' choice of high school program type–academic, vocational, or general. Since we have not yet covered the necessary tools for modelling such categorical outcomes, we will follow Faraway (2014, Chapter 16) and shift our focus slightly.

   (a) Model the math score as a function of the following five factors: gender, race, socioeconomic class, school type, and choice of high school program. Include all second-order interactions but no higher-order interactions. Fit the model in `R` and check its underlying assumptions. **Hint**: A model including all second order interactions among predictors `u`, `v`, `w`, and `z` can be fitted as

```r
lm(y ~ (u + v + w + z)^2)
```

(b) Perform backward model selection (e.g., using the `step` function) and write down your final model using the default identifiability constraints used by `R`. Do not forget to check that your final model does not violate any of the assumptions of the linear model.

(c) Briefly report your findings based on the model obtained in (b).

6. The dataset `teengamb` in the `faraway` package concerns a study of teenage gambling in Britain. To learn more about the variables in the dataset, please type `help(teengamb)`.

   (a) Fit a regression model with gambling expenditure as the response and sex, socioeconomic status, income, and verbal score as predictors. Then fit the same model using the square root of gambling expenditure as the response instead. Why should the second model be preferred?

   For the remaining questions, please use the square root of the expenditure as the response variable.

   (b) With all other predictors held constant, what is the estimated difference in the mean (square-root) expenditure on gambling for a male versus a female? Provide a 95% confidence interval for this difference.

   (c) Predict the gambling expenditure for a male with socioeconomic status, income, and verbal score all at their maximum values in the dataset, and construct a 95% prediction interval. Express your answer in the original units of expenditure.

   (d) Fit a model with income as the only predictor, and use an F-test to compare it with the full model from part (a). Clearly state the two models being compared. What does this tell you about the contribution of the remaining three predictors?

   (e) A simple model-selection approach is to consider all possible subsets of predictors. Assume that sex must be included in all models, and exclude interaction terms for simplicity. Fit all possible combinations of the remaining predictors. How stable is the estimated effect of sex across these models?

   (f) Which of the models fitted in part (e) would you choose as your final model, and on what basis? What conclusions would you draw from your selected model?